CLASSROOM TIME TRIALS & COMPETITION

StackCalc Calculator Applications

UIL-Style speed and accuracy proficiency tests aligned to maximum stack depth limits.

1. INVITATIONAL TIME TRIAL & SPEED SCORING RULES

Modeled after University Interscholastic League (UIL) Calculator Applications competitions, these timed trials challenge students to build rapid, instinctive fluency with Reverse Polish Notation (RPN). Each sprint tests numerical accuracy, operand sequencing, and strict stack depth consciousness.

CORRECT ANSWER	INCORRECT / ERROR	UNATTEMPTED / SKIP	SPEED BONUS
+5 Points	-2 Points	0 Points	+1 pt / rem. min

2. MAXIMUM STACK LEVEL DISCIPLINE

Competitors must respect the Max Stack Depth badge on each trial. Problems must be solved using only the allowed stack registers (Stack Level 1 = single register X; Stack Level 2 = registers X & Y; Stack Level 4 = full 4-register stack T-Z-Y-X). Solutions requiring deeper stacks are disqualified.

3. OFFICIAL COMPETITOR TIME TRIAL LOG SHEET

Trial	Topic / Stage	Max Stack	Time	Attempt	Correct	Errors	
T-1	Unit 1 Time Trial: Step Counter & Accumu	Level 1	5 min	___/10	___(+5)	___(-2)	_____
T-2	Unit 2 Time Trial: Part-Part-Whole & Sta	Level 2	5 min	___/10	___(+5)	___(-2)	_____
T-3	Unit 3 Time Trial: Ten-Frames & Compleme	Level 2	5 min	___/10	___(+5)	___(-2)	_____
T-4	Unit 4 Time Trial: Number Racks, Doubles	Level 2	5 min	___/10	___(+5)	___(-2)	_____
T-CUM_1	Cumulative Trial C1: Primary Stack Funda	Level 2	10 min	___/10	___(+5)	___(-2)	_____
T-5	Unit 5 Time Trial: 4-Register Stack & Fo	Level 4	10 min	___/10	___(+5)	___(-2)	_____
T-6	Unit 6 Time Trial: Fraction Studio & Mix	Level 4	10 min	___/10	___(+5)	___(-2)	_____
T-7	Unit 7 Time Trial: Postfix Evaluation &	Level 4	10 min	___/10	___(+5)	___(-2)	_____
T-8	Unit 8 Time Trial: Tactile Algebra & Qua	Level 4	10 min	___/10	___(+5)	___(-2)	_____
T-CUM_2	Cumulative Trial C2: Intermediate Stack	Level 4	15 min	___/10	___(+5)	___(-2)	_____
T-9	Unit 9 Time Trial: Trigonometry, Vectors	Level 4	10 min	___/10	___(+5)	___(-2)	_____
T-10	Unit 10 Time Trial: Exponential Growth &	Level 4	10 min	___/10	___(+5)	___(-2)	_____
T-11	Unit 11 Time Trial: Combinatorics, Stats	Level 4	10 min	___/10	___(+5)	___(-2)	_____
T-12	Unit 12 Time Trial: Digital Logic, Base	Level 4	10 min	___/10	___(+5)	___(-2)	_____
T-CUM_3	Cumulative Trial C3: Grand RPN Invitatio	Level 4	20 min	___/10	___(+5)	___(-2)	_____

Classroom Implementation

Run time trials as weekly sprint rounds. Time starts when students turn the page. All keystrokes must be entered on the physical StackCalc hardware or firmware simulator.

STUDENT CALCULATOR APPLICATIONS

Unit 1 Time Trial: Step Counter & Accumulator

Foundations of Counting, Cardinality & The One-More Step Counter • Stage: Ten-Frame Stage (4x2)

MAX STACK DEPTH: 1 | TIME LIMIT: 5 MIN | MAX SCORE: 50 PTS

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**1 + 1 + 1 (Count 3 steps)** Stack Hint: Single register accumulator	Keystroke Entry:	**Final Answer:**
2	**5 + 1 + 1 (Start at 5, step 2)** Stack Hint: Accumulate in X	Keystroke Entry:	**Final Answer:**
3	**Step count 4 items: start 0, step +1** Stack Hint: Repeat + key	Keystroke Entry:	**Final Answer:**
4	**Count 8 items with auto-lift** Stack Hint: Continuous + entry	Keystroke Entry:	**Final Answer:**
5	**Count down: start 6, step -1, step -1** Stack Hint: Subtract step	Keystroke Entry:	**Final Answer:**
6	**Clear and count: C then 1+1+1+1+1** Stack Hint: Clear before count	Keystroke Entry:	**Final Answer:**
7	**Count by 2s: 2 + 2 + 2** Stack Hint: Skip count by 2	Keystroke Entry:	**Final Answer:**
8	**Step count to ten: 7 + 1 + 1 + 1** Stack Hint: Reach upper frame	Keystroke Entry:	**Final Answer:**
9	**Zero set: 0 + 9** Stack Hint: Identity of zero	Keystroke Entry:	**Final Answer:**
10	**Count back from 10: 10 - 1 - 1 - 1** Stack Hint: Decrement from 10	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 1 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STUDENT CALCULATOR APPLICATIONS

Unit 2 Time Trial: Part-Part-Whole & Stack Ad

Part-Part-Whole Domino Decompositions via Stack Registers Y and X • Stage: Math Domino Studio (12-Frame Tray)

MAX STACK DEPTH: 2 | **TIME LIMIT: 5 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**Part 4, Part 3 -> Whole?** Stack Hint: Y=4, X=3, [+] combines	Keystroke Entry:	**Final Answer:**
2	**Part 5, Part 5 -> Whole?** Stack Hint: Ten bond	Keystroke Entry:	**Final Answer:**
3	**Whole 9, Part 6 -> Missing Part?** Stack Hint: Decompose whole 9	Keystroke Entry:	**Final Answer:**
4	**Whole 10, Part 4 -> Missing Part?** Stack Hint: Decompose whole 10	Keystroke Entry:	**Final Answer:**
5	**Part 8, Part 7 -> Whole?** Stack Hint: Sum past ten	Keystroke Entry:	**Final Answer:**
6	**Whole 14, Part 8 -> Missing Part?** Stack Hint: Decompose 14	Keystroke Entry:	**Final Answer:**
7	**Part 6, Part 7 -> Whole?** Stack Hint: Part combination	Keystroke Entry:	**Final Answer:**
8	**Whole 18, Part 9 -> Missing Part?** Stack Hint: Double decomposition	Keystroke Entry:	**Final Answer:**
9	**Part 12, Part 8 -> Whole?** Stack Hint: Full twenty bond	Keystroke Entry:	**Final Answer:**
10	**Whole 25, Part 11 -> Missing Part?** Stack Hint: Two-digit decompose	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 2 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STUDENT CALCULATOR APPLICATIONS

Unit 3 Time Trial: Ten-Frames & Complements t

5-Structure, Complements & Bridging Through 10 • Stage: Ten-Frame Stage (4x2)

MAX STACK DEPTH: 2 | TIME LIMIT: 5 MIN | MAX SCORE: 50 PTS

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**Complement of 7 to 10** Stack Hint: 10 in Y, 7 in X	Keystroke Entry:	**Final Answer:**
2	**Complement of 4 to 10** Stack Hint: 10 in Y, 4 in X	Keystroke Entry:	**Final Answer:**
3	**Bridge: 8 + 5 = 8 + 2 + 3** Stack Hint: Bridge through 10	Keystroke Entry:	**Final Answer:**
4	**Bridge: 9 + 6 = 9 + 1 + 5** Stack Hint: Bridge through 10	Keystroke Entry:	**Final Answer:**
5	**Bridge: 7 + 6 = 7 + 3 + 3** Stack Hint: Bridge through 10	Keystroke Entry:	**Final Answer:**
6	**Complement of 1 to 10** Stack Hint: 10 in Y, 1 in X	Keystroke Entry:	**Final Answer:**
7	**Double frame: 20 - 7** Stack Hint: Subtract across 2 frames	Keystroke Entry:	**Final Answer:**
8	**Bridge: 8 + 7 = 8 + 2 + 5** Stack Hint: Bridge through 10	Keystroke Entry:	**Final Answer:**
9	**Place value: 10 + 6** Stack Hint: Teen number formation	Keystroke Entry:	**Final Answer:**
10	**Frame difference: 10 - 2** Stack Hint: Ten complement	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 2 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STUDENT CALCULATOR APPLICATIONS

Unit 4 Time Trial: Number Racks, Doubles & Gr

5-Structure, Automatic Lift Duplication & Early Multiplication • Stage: Number Rack (4x2)

MAX STACK DEPTH: 2 | TIME LIMIT: 5 MIN | MAX SCORE: 50 PTS

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**Double 6: 6 ENTER +** Stack Hint: ENTER duplicates to Y	Keystroke Entry:	**Final Answer:**
2	**Double 8: 8 ENTER +** Stack Hint: ENTER duplicates to Y	Keystroke Entry:	**Final Answer:**
3	**Near-Double: 7 + 8** Stack Hint: Double 7 plus 1	Keystroke Entry:	**Final Answer:**
4	**Near-Double: 6 + 7** Stack Hint: Double 6 plus 1	Keystroke Entry:	**Final Answer:**
5	**Double 9: 9 ENTER +** Stack Hint: ENTER duplicates to Y	Keystroke Entry:	**Final Answer:**
6	**Near-Double: 8 + 9** Stack Hint: Double 8 plus 1	Keystroke Entry:	**Final Answer:**
7	**Equal groups: 3 groups of 4** Stack Hint: Multiply group size	Keystroke Entry:	**Final Answer:**
8	**Equal groups: 4 groups of 5** Stack Hint: Multiply group size	Keystroke Entry:	**Final Answer:**
9	**Double 12: 12 ENTER +** Stack Hint: ENTER duplicates to Y	Keystroke Entry:	**Final Answer:**
10	**Near-Double: 9 + 10** Stack Hint: Double 9 plus 1	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 2 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STUDENT CALCULATOR APPLICATIONS

Cumulative Trial C1: Primary Stack Fundamenta

Comprehensive Time Trial across Units 1–4 (Max Stack Depth 2) • Stage: Gridfinity 4x4 Primary Desk Dock

MAX STACK DEPTH: 2 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**Step count 7 items from 1** Stack Hint: Accumulate in X	Keystroke Entry:	**Final Answer:**
2	**Number bond: 9 and 7 -> Whole?** Stack Hint: Y=9, X=7	Keystroke Entry:	**Final Answer:**
3	**Decompose: Whole 17, Part 8** Stack Hint: 17 in Y, 8 in X	Keystroke Entry:	**Final Answer:**
4	**Complement of 3 to 10** Stack Hint: Ten complement	Keystroke Entry:	**Final Answer:**
5	**Double 14: 14 ENTER +** Stack Hint: Duplicate to Y	Keystroke Entry:	**Final Answer:**
6	**Near-double: 11 + 12** Stack Hint: Double 11 plus 1	Keystroke Entry:	**Final Answer:**
7	**Bridge through 10: 9 + 8** Stack Hint: Bridge with frame	Keystroke Entry:	**Final Answer:**
8	**Equal groups: 6 groups of 5** Stack Hint: 5 in Y, 6 in X	Keystroke Entry:	**Final Answer:**
9	**Decompose: Whole 30, Part 12** Stack Hint: Missing part	Keystroke Entry:	**Final Answer:**
10	**Combined: Double 8 then add 5** Stack Hint: Chained operations	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 2 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STUDENT CALCULATOR APPLICATIONS

Unit 5 Time Trial: 4-Register Stack & Four Op

T-Z-Y-X Flow, Automatic Lift & Parenthesis-Free Arithmetic • Stage: SC-32 Hardware LCD Stack & Place Value Studio

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	(18 - 6) × 3 Stack Hint: Auto-lift pushes 12 to Y	Keystroke Entry:	Final Answer:
2	(24 + 16) ÷ 8 Stack Hint: Sum in Y, divide by 8	Keystroke Entry:	Final Answer:
3	45 ÷ (12 - 3) Stack Hint: Push 45 to Y, sub in X	Keystroke Entry:	Final Answer:
4	(15 × 4) + (18 ÷ 2) Stack Hint: Two branch accumulation	Keystroke Entry:	Final Answer:
5	(50 - 14) ÷ (3 + 6) Stack Hint: Stack holds numerator in Y	Keystroke Entry:	Final Answer:
6	12 × 5 - 8 × 4 Stack Hint: Two branch subtraction	Keystroke Entry:	Final Answer:
7	((20 - 8) × 4) ÷ 6 Stack Hint: Chained pipeline	Keystroke Entry:	Final Answer:
8	72 ÷ (4 + 5) × 3 Stack Hint: Parenthesis-free stack	Keystroke Entry:	Final Answer:
9	100 - (16 × 5 - 12) Stack Hint: Deep stack evaluate	Keystroke Entry:	Final Answer:
10	(8 + 4) × (9 - 3) ÷ 12 Stack Hint: Multi-step evaluate	Keystroke Entry:	Final Answer:

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STUDENT CALCULATOR APPLICATIONS

Unit 6 Time Trial: Fraction Studio & Mixed Nu

Two-Dot Notation, Reciprocals & Height Equivalence • Stage: Fraction Studio (4x2)

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**1/2 + 1/4 (Two-dot entry)** Stack Hint: Decimal equivalent 0.75	Keystroke Entry:	**Final Answer:**
2	**3/4 - 1/2** Stack Hint: Decimal equivalent 0.25	Keystroke Entry:	**Final Answer:**
3	**Mixed: 2 1/2 + 1 1/4** Stack Hint: Mixed number addition	Keystroke Entry:	**Final Answer:**
4	**Reciprocal of 4/5: .4.5 [1/x]** Stack Hint: Reciprocal key	Keystroke Entry:	**Final Answer:**
5	**3/8 + 1/4** Stack Hint: Three eighths plus fourth	Keystroke Entry:	**Final Answer:**
6	**2/3 × 3/4** Stack Hint: Fraction multiplication	Keystroke Entry:	**Final Answer:**
7	**3/4 ÷ 1/2** Stack Hint: Fraction division	Keystroke Entry:	**Final Answer:**
8	**1 1/2 × 4** Stack Hint: Mixed scaling	Keystroke Entry:	**Final Answer:**
9	**5 - 2 3/4** Stack Hint: Subtract mixed number	Keystroke Entry:	**Final Answer:**
10	**(1/3 + 1/6) × 12** Stack Hint: Combined fraction pipeline	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STUDENT CALCULATOR APPLICATIONS

Unit 7 Time Trial: Postfix Evaluation & Chain

Postfix Evaluation, Stack Registers & Parenthesis Elimination • Stage: Interlocking Cubes & SC-32 Hardware Stack

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	(8 + 2) × (9 - 4) Stack Hint: Left-Right-Multiply	Keystroke Entry:	**Final Answer:**
2	((14 - 6) × 3) + (24 ÷ 4) Stack Hint: Two subtree combine	Keystroke Entry:	**Final Answer:**
3	(30 - (4 × 5)) × (12 ÷ (2 + 4)) Stack Hint: Deep nested subtrees	Keystroke Entry:	**Final Answer:**
4	(5 + 3) × (8 - 2) ÷ (6 - 2) Stack Hint: Three branch traversal	Keystroke Entry:	**Final Answer:**
5	((18 ÷ 3) + 4) × ((15 - 7) ÷ 2) Stack Hint: Postorder multiplication	Keystroke Entry:	**Final Answer:**
6	(10 + 5 × 2) ÷ (12 - 2 × 4) Stack Hint: Eliminating PEMDAS ambiguity	Keystroke Entry:	**Final Answer:**
7	(25 - 5 × 3) × (4 + 6 ÷ 2) Stack Hint: Nested tree traversal	Keystroke Entry:	**Final Answer:**
8	((7 + 3) × 4 - 8) ÷ 4 Stack Hint: Chained operations	Keystroke Entry:	**Final Answer:**
9	(48 ÷ (2 + 6)) × (15 - 3 × 3) Stack Hint: Tree sub-expression	Keystroke Entry:	**Final Answer:**
10	(16 - (4 + 2)) × (18 ÷ (3 × 2)) Stack Hint: Stack tree depth 4	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STUDENT CALCULATOR APPLICATIONS

Unit 8 Time Trial: Tactile Algebra & Quadrati

Area Modeling, Zero-Pair Trinomials & [EQN] Solver • Stage: Algebra Studio & Factoring Stage

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**Factor $x^2 + 5x + 6$ via [EQN]** Stack Hint: Roots -2, -3; factors (x+2)(x+3)	Keystroke Entry:	**Final Answer:**
2	**Factor $x^2 + 7x + 12$ via [EQN]** Stack Hint: Roots -3, -4; factors (x+3)(x+4)	Keystroke Entry:	**Final Answer:**
3	**Factor $x^2 + 8x + 15$ via [EQN]** Stack Hint: Roots -3, -5; factors (x+3)(x+5)	Keystroke Entry:	**Final Answer:**
4	**Factor $x^2 - x - 6$ via [EQN]** Stack Hint: Roots 3, -2; factors (x-3)(x+2)	Keystroke Entry:	**Final Answer:**
5	**Factor $x^2 - 5x + 6$ via [EQN]** Stack Hint: Roots 2, 3; factors (x-2)(x-3)	Keystroke Entry:	**Final Answer:**
6	**Difference of squares: $x^2 - 9$** Stack Hint: Roots 3, -3; factors (x-3)(x+3)	Keystroke Entry:	**Final Answer:**
7	**Difference of squares: $x^2 - 16$** Stack Hint: Roots 4, -4; factors (x-4)(x+4)	Keystroke Entry:	**Final Answer:**
8	**Polynomial at x=3: $2x^2 + 4x - 5$** Stack Hint: Quadratic evaluation	Keystroke Entry:	**Final Answer:**
9	**Factor $2x^2 + 5x + 2$ via [EQN]** Stack Hint: Roots -0.5, -2; (2x+1)(x+2)	Keystroke Entry:	**Final Answer:**
10	**Double root: $x^2 + 10x + 25$** Stack Hint: Double root -5; $(x+5)^2$	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STUDENT CALCULATOR APPLICATIONS

Cumulative Trial C2: Intermediate Stack Maste

Comprehensive Time Trial across Units 1–8 (Max Stack Depth 4) • Stage: SC-32 Hardware LCD Stack & Manipulative Lab

MAX STACK DEPTH: 4 | **TIME LIMIT: 15 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**(32 - 14) × 5 ÷ 6** Stack Hint: Auto-lift chaining	Keystroke Entry:	**Final Answer:**
2	**3/4 + 1/2 × 1/4 (order of ops)** Stack Hint: Fractions on stack	Keystroke Entry:	**Final Answer:**
3	**Distance: P1(3, 4) to P2(9, 12)** Stack Hint: Pythagorean distance	Keystroke Entry:	**Final Answer:**
4	**Mixed: 3 1/2 - 1 3/4** Stack Hint: Two-dot fraction subtraction	Keystroke Entry:	**Final Answer:**
5	**Postfix evaluation: ((15+5)×(8-3))÷10** Stack Hint: Postfix evaluation	Keystroke Entry:	**Final Answer:**
6	**Double 18 then sub 9** Stack Hint: Duplicate to Y	Keystroke Entry:	**Final Answer:**
7	**Slope: P1(-1, -2) to P2(3, 10)** Stack Hint: Negative coordinates	Keystroke Entry:	**Final Answer:**
8	**Four operations: 80 - 15 × 4 + 12** Stack Hint: RPN stack evaluation	Keystroke Entry:	**Final Answer:**
9	**Reciprocal sum: 1/(1/4 + 1/6)** Stack Hint: Reciprocal pipeline	Keystroke Entry:	**Final Answer:**
10	**Distance from origin to (12, 16)** Stack Hint: Radial distance	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STUDENT CALCULATOR APPLICATIONS

Unit 9 Time Trial: Trigonometry, Vectors & Po

Right Triangles, Polar Conversions & Vector Decomposition • Stage: Unit Circle Trigonometry Volvelle

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**sin(30°)** Stack Hint: Basic sine ratio	Keystroke Entry:	**Final Answer:**
2	**cos(60°)** Stack Hint: Basic cosine ratio	Keystroke Entry:	**Final Answer:**
3	**tan(45°)** Stack Hint: Tangent ratio	Keystroke Entry:	**Final Answer:**
4	**Hypotenuse 10, angle 30° -> Opposite (y)** Stack Hint: Opposite = h × sin(□)	Keystroke Entry:	**Final Answer:**
5	**Adjacent (x): Hypotenuse 12, angle 60°** Stack Hint: Adjacent = h × cos(□)	Keystroke Entry:	**Final Answer:**
6	**Polar to Rect: r=10, □=0° -> x** Stack Hint: Polar conversion key	Keystroke Entry:	**Final Answer:**
7	**Rect to Polar: x=3, y=4 -> magnitude r** Stack Hint: Vector magnitude	Keystroke Entry:	**Final Answer:**
8	**Rect to Polar: x=5, y=12 -> magnitude r** Stack Hint: Vector magnitude	Keystroke Entry:	**Final Answer:**
9	**Vector magnitude: components 8 and 6** Stack Hint: Polar magnitude	Keystroke Entry:	**Final Answer:**
10	**asin(0.5) in degrees** Stack Hint: Inverse sine	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STUDENT CALCULATOR APPLICATIONS

Unit 10 Time Trial: Exponential Growth & Loga

Natural Base e, Continuous Compounding & Doubling Time • Stage: Fraction Studio & SC-32 Hardware LCD Stack

MAX STACK DEPTH: 4 | TIME LIMIT: 10 MIN | MAX SCORE: 50 PTS

Competitor: _____ Date: _____ Elapsed: ____:____ Score: ____ / 50

1	**2^5 (Power key)** Stack Hint: y to power x	Keystroke Entry:	Final Answer:
2	**3^4 (Power key)** Stack Hint: y to power x	Keystroke Entry:	Final Answer:
3	**ln(e)** Stack Hint: Natural log inverse	Keystroke Entry:	Final Answer:
4	**e^0** Stack Hint: Base e exponent	Keystroke Entry:	Final Answer:
5	**log10(1000)** Stack Hint: Common log	Keystroke Entry:	Final Answer:
6	**Compound: 1000 × (1.05)^2** Stack Hint: Annual compounding	Keystroke Entry:	Final Answer:
7	**Doubling time: ln(2) ÷ 0.07** Stack Hint: Continuous doubling rule	Keystroke Entry:	Final Answer:
8	**Half-life: 100 × (0.5)^3** Stack Hint: Decay model	Keystroke Entry:	Final Answer:
9	**Continuous: 500 × e^(0.04 × 5)** Stack Hint: P e^(rt) formula	Keystroke Entry:	Final Answer:
10	**2^10** Stack Hint: Powers of two	Keystroke Entry:	Final Answer:

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STUDENT CALCULATOR APPLICATIONS

Unit 11 Time Trial: Combinatorics, Stats & Pr

Factorials, Permutations, Combinations & [Σ+] Accumulation • Stage: Probability Dice & Modular Spinner

MAX STACK DEPTH: 4 | TIME LIMIT: 10 MIN | MAX SCORE: 50 PTS

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**5! (Factorial)** Stack Hint: x factorial key	Keystroke Entry:	**Final Answer:**
2	**6! (Factorial)** Stack Hint: x factorial key	Keystroke Entry:	**Final Answer:**
3	**Permutations P(5, 2) = 5! / 3!** Stack Hint: Stack permutation	Keystroke Entry:	**Final Answer:**
4	**Permutations P(6, 3) = 6! / 3!** Stack Hint: Stack permutation	Keystroke Entry:	**Final Answer:**
5	**Combinations C(5, 2) = 5! / (2! × 3!)** Stack Hint: n choose r	Keystroke Entry:	**Final Answer:**
6	**Combinations C(6, 2) = 6! / (2! × 4!)** Stack Hint: n choose r	Keystroke Entry:	**Final Answer:**
7	**Combinations C(7, 3) = 7! / (3! × 4!)** Stack Hint: n choose r	Keystroke Entry:	**Final Answer:**
8	**Series sum 1+2+3+4+5 via [Σ+]** Stack Hint: Statistical summation	Keystroke Entry:	**Final Answer:**
9	**4! (Factorial)** Stack Hint: x factorial key	Keystroke Entry:	**Final Answer:**
10	**Probability: 1 ÷ C(6, 2)** Stack Hint: Reciprocal combination	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STUDENT CALCULATOR APPLICATIONS

Unit 12 Time Trial: Digital Logic, Base Conve

HEX/DEC/OCT/BIN Radix Modes & Bitwise AND, OR, XOR Operations • Stage: Ten-Frame Stage (4x2)

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**DEC 255 to HEX** Stack Hint: Radix mode cycle	Keystroke Entry:	**Final Answer:**
2	**HEX 10 to DEC** Stack Hint: Hex to decimal	Keystroke Entry:	**Final Answer:**
3	**DEC 15 to BIN** Stack Hint: Dec to binary	Keystroke Entry:	**Final Answer:**
4	**BIN 1010 to DEC** Stack Hint: Bin to decimal	Keystroke Entry:	**Final Answer:**
5	**Bitwise AND: 12 AND 10** Stack Hint: 1100 AND 1010 = 1000	Keystroke Entry:	**Final Answer:**
6	**Bitwise OR: 12 OR 10** Stack Hint: 1100 OR 1010 = 1110	Keystroke Entry:	**Final Answer:**
7	**Bitwise XOR: 12 XOR 10** Stack Hint: 1100 XOR 1010 = 0110	Keystroke Entry:	**Final Answer:**
8	**HEX 20 to DEC** Stack Hint: Hex 20 is 32	Keystroke Entry:	**Final Answer:**
9	**DEC 64 to OCT** Stack Hint: Dec 64 is octal 100	Keystroke Entry:	**Final Answer:**
10	**Bitwise AND: 255 AND 15** Stack Hint: Low nybble mask	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STUDENT CALCULATOR APPLICATIONS

Cumulative Trial C3: Grand RPN Invitational C

Master UIL-Style Time Trial across Units 1–12 (Complete Architecture) • Stage: Master Gridfinity Classroom Ecosystem

MAX STACK DEPTH: 4 | **TIME LIMIT: 20 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**Compound pipeline: 250 × (1 + .06/12)^24** Stack Hint: TVM compounding	Keystroke Entry:	**Final Answer:**
2	**Pythagorean distance: P1(-3, 5) to P2(5, -1(** Stack Hint: Distance pipeline triple	Keystroke Entry:	**Final Answer:**
3	**Polar vector: 12 cos(45°) + 5 sin(45°)** Stack Hint: Vector projection	Keystroke Entry:	**Final Answer:**
4	**Fractions: 3 1/3 ÷ 1 1/4 + 2/5** Stack Hint: Mixed operations	Keystroke Entry:	**Final Answer:**
5	**Combinations: C(8, 3) = 8! / (3! × 5!)** Stack Hint: Lottery combination	Keystroke Entry:	**Final Answer:**
6	**Exponential: 1500 × e^(0.05 × 10)** Stack Hint: Continuous growth	Keystroke Entry:	**Final Answer:**
7	**Postfix evaluation: ((35-7)×4)÷(14+2)** Stack Hint: Parenthesis elimination	Keystroke Entry:	**Final Answer:**
8	**Bitmasking: 192 AND 240 (Subnet mask)** Stack Hint: Network bitwise logic	Keystroke Entry:	**Final Answer:**
9	**Double & near-double sum: (6 ENTER +) + (8** Stack Hint: Stack auto-lift chain	Keystroke Entry:	**Final Answer:**
10	**Radix conversion: DEC 4095 to HEX** Stack Hint: Maximum 12-bit hex	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

TEACHER COMPETITION SOLUTION GUIDE

Official Solutions & Keystrokes — Part 1

Answer keys, required numerical results, and canonical keystroke sequences for Trials 1 to 6 and C1.

Unit 1 Time Trial: Step Counter (Stack 1)

1. Ans: 3 — [1] [ENTER] [1] [+] [1] [+]
2. Ans: 7 — [5] [ENTER] [1] [+] [1] [+]
3. Ans: 4 — [1] [ENTER] [+] [+] [+]
4. Ans: 8 — [1] [ENTER] [+] [+] [+] [+]
5. Ans: 4 — [6] [ENTER] [1] [-] [1] [-]
6. Ans: 5 — [C] [1] [ENTER] [+] [+] [+]
7. Ans: 6 — [2] [ENTER] [2] [+] [2] [+]
8. Ans: 10 — [7] [ENTER] [1] [+] [1] [+]
9. Ans: 9 — [0] [ENTER] [9] [+]
10. Ans: 7 — [10] [ENTER] [1] [-] [1] [-]

Unit 2 Time Trial: Part-Part-Who (Stack 2)

1. Ans: 7 — [4] [ENTER] [3] [+]
2. Ans: 10 — [5] [ENTER] [5] [+]
3. Ans: 3 — [9] [ENTER] [6] [-]
4. Ans: 6 — [10] [ENTER] [4] [-]
5. Ans: 15 — [8] [ENTER] [7] [+]
6. Ans: 6 — [14] [ENTER] [8] [-]
7. Ans: 13 — [6] [ENTER] [7] [+]
8. Ans: 9 — [18] [ENTER] [9] [-]
9. Ans: 20 — [12] [ENTER] [8] [+]
10. Ans: 14 — [25] [ENTER] [11] [-]

Unit 3 Time Trial: Ten-Frames & (Stack 2)

1. Ans: 3 — [10] [ENTER] [7] [-]
2. Ans: 6 — [10] [ENTER] [4] [-]
3. Ans: 13 — [8] [ENTER] [2] [+] [3] [+]
4. Ans: 15 — [9] [ENTER] [1] [+] [5] [+]
5. Ans: 13 — [7] [ENTER] [3] [+] [3] [+]
6. Ans: 9 — [10] [ENTER] [1] [-]
7. Ans: 13 — [20] [ENTER] [7] [-]
8. Ans: 15 — [8] [ENTER] [2] [+] [5] [+]
9. Ans: 16 — [10] [ENTER] [6] [+]
10. Ans: 8 — [10] [ENTER] [2] [-]

Unit 4 Time Trial: Number Racks, (Stack 2)

1. Ans: 12 — [6] [ENTER] [+]
2. Ans: 16 — [8] [ENTER] [+]
3. Ans: 15 — [7] [ENTER] [+] [1] [+]
4. Ans: 13 — [6] [ENTER] [+] [1] [+]
5. Ans: 18 — [9] [ENTER] [+]
6. Ans: 17 — [8] [ENTER] [+] [1] [+]
7. Ans: 12 — [4] [ENTER] [3] [×]
8. Ans: 20 — [5] [ENTER] [4] [×]
9. Ans: 24 — [12] [ENTER] [+]
10. Ans: 19 — [9] [ENTER] [+] [1] [+]

Cumulative Trial C1: Primary Sta (Stack 2)

1. Ans: 7 — [1] [ENTER] [+] [+] [+] [+]
2. Ans: 16 — [9] [ENTER] [7] [+]
3. Ans: 9 — [17] [ENTER] [8] [-]
4. Ans: 7 — [10] [ENTER] [3] [-]
5. Ans: 28 — [14] [ENTER] [+]
6. Ans: 23 — [11] [ENTER] [+] [1] [+]
7. Ans: 17 — [9] [ENTER] [1] [+] [7] [+]
8. Ans: 30 — [5] [ENTER] [6] [×]
9. Ans: 18 — [30] [ENTER] [12] [-]
10. Ans: 21 — [8] [ENTER] [+] [5] [+]

Unit 5 Time Trial: 4-Register St (Stack 4)

1. Ans: 36 — [18] [ENTER] [6] [-] [3] [×]
2. Ans: 5 — [24] [ENTER] [16] [+] [8] [÷]
3. Ans: 5 — [45] [ENTER] [12] [ENTER] [3] [-]
4. Ans: 69 — [15] [ENTER] [4] [×] [18] [ENTER]
5. Ans: 4 — [50] [ENTER] [14] [-] [3] [ENTER]
6. Ans: 28 — [12] [ENTER] [5] [×] [8] [ENTER]
7. Ans: 8 — [20] [ENTER] [8] [-] [4] [×]
8. Ans: 24 — [72] [ENTER] [4] [ENTER] [5] [+]
9. Ans: 32 — [100] [ENTER] [16] [ENTER] [5] [×]
10. Ans: 6 — [8] [ENTER] [4] [+] [9] [ENTER]

Unit 6 Time Trial: Fraction Stud (Stack 4)

1. Ans: 0.75 — [.1.2] [ENTER] [.1.4] [+]
2. Ans: 0.25 — [.3.4] [ENTER] [.1.2] [-]
3. Ans: 3.75 — [2.1.2] [ENTER] [1.1.4] [+]
4. Ans: 1.25 — [.4.5] [1/x]
5. Ans: 0.625 — [.3.8] [ENTER] [.1.4] [+]
6. Ans: 0.5 — [.2.3] [ENTER] [.3.4] [×]
7. Ans: 1.5 — [.3.4] [ENTER] [.1.2] [÷]
8. Ans: 6 — [1.1.2] [ENTER] [4] [×]
9. Ans: 2.25 — [5] [ENTER] [2.3.4] [-]
10. Ans: 6 — [.1.3] [ENTER] [.1.6] [+] [12] [×]

Scoring Adjudication

Accept equivalent decimal fractions (e.g. 0.75 or 3/4). If a student pressed superfluous ENTER keys exceeding the maximum stack depth, deduct 2 points for stack overflow violation.

TEACHER COMPETITION SOLUTION GUIDE

Official Solutions & Keystrokes — Part 2

Answer keys, required numerical results, and canonical keystroke sequences for Trials 7 to 12, C2 & C3.

Unit 7 Time Trial: Postfix Evalu (Stack 4)

1. Ans: 50 — [8] [ENTER] [2] [+] [9] [ENTER]
2. Ans: 30 — [14] [ENTER] [6] [-] [3] [×]
3. Ans: 20 — [30] [ENTER] [4] [ENTER] [5] [×]
4. Ans: 12 — [5] [ENTER] [3] [+] [8] [ENTER]
5. Ans: 40 — [18] [ENTER] [3] [÷] [4] [+]
6. Ans: 5 — [10] [ENTER] [5] [ENTER] [2] [×]
7. Ans: 70 — [25] [ENTER] [5] [ENTER] [3] [×]
8. Ans: 8 — [7] [ENTER] [3] [+] [4] [×]
9. Ans: 36 — [48] [ENTER] [2] [ENTER] [6] [+]
10. Ans: 30 — [16] [ENTER] [4] [ENTER] [2] [+]

Unit 8 Time Trial: Tactile Algeb (Stack 4)

1. Ans: -2 — [1] [ENTER] [5] [ENTER] [6] [EQN]
2. Ans: -3 — [1] [ENTER] [7] [ENTER] [12] [EQN]
3. Ans: -3 — [1] [ENTER] [8] [ENTER] [15] [EQN]
4. Ans: 3 — [1] [ENTER] [1] [+/-] [ENTER] [6]
5. Ans: 2 — [1] [ENTER] [5] [+/-] [ENTER] [6]
6. Ans: 3 — [1] [ENTER] [0] [ENTER] [9] [+/-]
7. Ans: 4 — [1] [ENTER] [0] [ENTER] [16] [+/-]
8. Ans: 25 — [3] [ENTER] [x²] [2] [×] [3]
9. Ans: -0.5 — [2] [ENTER] [5] [ENTER] [2] [EQN]
10. Ans: -5 — [1] [ENTER] [10] [ENTER] [25] [EQN

Cumulative Trial C2: Intermediat (Stack 4)

1. Ans: 15 — [32] [ENTER] [14] [-] [5] [×]
2. Ans: 0.875 — [.1.2] [ENTER] [.1.4] [×] [.3.4] [
3. Ans: 10 — [9] [ENTER] [3] [-] [x²] [12]
4. Ans: 1.75 — [3.1.2] [ENTER] [1.3.4] [-]
5. Ans: 10 — [15] [ENTER] [5] [+] [8] [ENTER]
6. Ans: 27 — [18] [ENTER] [+] [9] [-]
7. Ans: 3 — [10] [ENTER] [2] [+/-] [-] [3]
8. Ans: 32 — [80] [ENTER] [15] [ENTER] [4] [×]
9. Ans: 2.4 — [.1.4] [ENTER] [.1.6] [+] [1/x]
10. Ans: 20 — [12] [x²] [16] [x²] [+] [√☒]

Unit 9 Time Trial: Trigonometry, (Stack 4)

1. Ans: 0.5 — [30] [SIN]
2. Ans: 0.5 — [60] [COS]
3. Ans: 1 — [45] [TAN]
4. Ans: 5 — [10] [ENTER] [30] [SIN] [×]
5. Ans: 6 — [12] [ENTER] [60] [COS] [×]
6. Ans: 10 — [10] [ENTER] [0] [yellow] [☒y,x]
7. Ans: 5 — [3] [ENTER] [4] [yellow] [☒☒,r]
8. Ans: 13 — [5] [ENTER] [12] [yellow] [☒☒,r]
9. Ans: 10 — [8] [ENTER] [6] [yellow] [☒☒,r]
10. Ans: 30 — [.5] [yellow] [ASIN]

Unit 10 Time Trial: Exponential (Stack 4)

1. Ans: 32 — [2] [ENTER] [5] [y☒]
2. Ans: 81 — [3] [ENTER] [4] [y☒]
3. Ans: 1 — [1] [e☒] [LN]
4. Ans: 1 — [0] [e☒]
5. Ans: 3 — [1000] [yellow] [LOG]
6. Ans: 1102.5 — [1000] [ENTER] [1.05] [ENTER] [2]
7. Ans: 9.9021 — [2] [LN] [.07] [÷]
8. Ans: 12.5 — [100] [ENTER] [.5] [ENTER] [3] [y☒
9. Ans: 610.7014 — [.04] [ENTER] [5] [×] [e☒] [500]
10. Ans: 1024 — [2] [ENTER] [10] [y☒]

Unit 11 Time Trial: Combinatoric (Stack 4)

1. Ans: 120 — [5] [yellow] [x!]
2. Ans: 720 — [6] [yellow] [x!]
3. Ans: 20 — [5] [yellow] [x!] [3] [yellow] [x!
4. Ans: 120 — [6] [yellow] [x!] [3] [yellow] [x!
5. Ans: 10 — [5] [yellow] [x!] [2] [yellow] [x!
6. Ans: 15 — [6] [yellow] [x!] [2] [yellow] [x!
7. Ans: 35 — [7] [yellow] [x!] [3] [yellow] [x!
8. Ans: 15 — [1] [☒+] [2] [☒+] [3] [☒+]
9. Ans: 24 — [4] [yellow] [x!]
10. Ans: 0.0667 — [6] [yellow] [x!] [2] [yellow] [x!

Unit 12 Time Trial: Digital Logi (Stack 4)

1. Ans: FF — [255] [yellow] [BASE] [HEX]
2. Ans: 16 — [HEX] [10] [yellow] [BASE] [DEC]
3. Ans: 1111 — [15] [yellow] [BASE] [BIN]
4. Ans: 10 — [BIN] [1010] [yellow] [BASE] [DEC]
5. Ans: 8 — [12] [ENTER] [10] [AND]
6. Ans: 14 — [12] [ENTER] [10] [OR]
7. Ans: 6 — [12] [ENTER] [10] [XOR]
8. Ans: 32 — [HEX] [20] [yellow] [BASE] [DEC]
9. Ans: 100 — [64] [yellow] [BASE] [OCT]
10. Ans: 15 — [255] [ENTER] [15] [AND]

Cumulative Trial C3: Grand RPN I (Stack 4)

1. Ans: 281.79 — [.06] [ENTER] [12] [÷] [1] [+]
2. Ans: 17 — [5] [ENTER] [3] [+/-] [-] [x²]
3. Ans: 12.0208 — [12] [ENTER] [45] [COS] [×] [5]
4. Ans: 3.0667 — [3.1.3] [ENTER] [1.1.4] [÷] [.2.5]
5. Ans: 56 — [8] [yellow] [x!] [3] [yellow] [x!
6. Ans: 2473.08 — [.05] [ENTER] [10] [×] [e☒] [1500]
7. Ans: 7 — [35] [ENTER] [7] [-] [4] [×]
8. Ans: 192 — [192] [ENTER] [240] [AND]
9. Ans: 29 — [6] [ENTER] [+] [8] [ENTER] [+]
10. Ans: FFF — [4095] [yellow] [BASE] [HEX]

Scoring Adjudication

Accept equivalent scientific notation for large magnitudes (e.g. 1024 or 1.024e3). All answers match canonical RPNCore engine execution.